ON SHAKY GROUNDS

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Earthquake Prediction and Simulation System using the concepts of Machine Learning

V Semester

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Project Synopsis

Abstract:

Earthquakes pose a massive and unpredictable risk to global infrastructure and human life. This project proposes an end-to-end machine learning system for analysing seismic patterns, predicting earthquake severity, and interactively simulating geographic risks. Utilizing a comprehensive dataset sourced from the United States Geological Survey (USGS) containing over thousands of recorded seismic events, the system predicts the likely magnitude and severity of future earthquakes based on geographic, temporal, and instrumental factors. The problem is formulated as a multi-target forecasting and classification task. Data preprocessing includes missing-value treatment, spatial bounding box encoding, and feature interactions (such as depth-to-severity decay). Ensemble models, specifically Random Forest and XGBoost, are deployed for magnitude prediction (regression) and severity classification. Additionally, unsupervised K-Means clustering is utilized for identifying natural geological risk zones. Finally, the selected models are distilled into a heuristic simulation engine integrated into a sophisticated frontend architecture. The final product includes an interactive 3D WebGL globe, real-time USGS data ingestion, and a dynamic prediction simulator allowing researchers to test 'what-if' scenarios by varying fault mechanisms, local soil conditions, and proximity to population centres. By uniting predictive analytics with an intuitive, explorable interface, the system aims to make sophisticated seismic risk assessment accessible not only to domain experts but also to students, planners, and members of the public.

Keywords— *Earthquake Prediction; Machine Learning; XGBoost; Random Forest; Seismic Hazard Analysis; WebGL Visualization; Time-Series Analysis; K-Means Clustering; Real-time API Integration.*

I. INTRODUCTION

Earthquakes remain among the most destructive and least predictable natural hazards faced by modern society. Unlike hurricanes or floods, which typically provide days of advance warning through atmospheric monitoring, seismic events can strike with only seconds of precursory signal, leaving little time for evacuation or emergency response. The resulting damage to infrastructure, disruption to economies, and loss of human life underscore the urgent need for tools that can characterize seismic risk before, during, and after an event occurs.

Traditional approaches to seismic hazard assessment have relied heavily on physics-based models and statistical relationships such as the Gutenberg-Richter law, which describe the long-term frequency of earthquakes at a given magnitude but offer limited insight into localized, short-term risk. The growing availability of large, well-curated seismological datasets — most notably from the United States Geological Survey (USGS) Earthquake Hazards Program — has opened the door to data-driven approaches capable of learning complex, nonlinear relationships between geographic, temporal, and instrumental variables that classical models struggle to capture.

This project, titled “On Shaky Grounds,” proposes an end-to-end machine learning pipeline that combines supervised regression and classification techniques with unsupervised clustering to forecast earthquake magnitude, classify surface severity, and identify natural tectonic risk zones. The learned models are further distilled into a lightweight, client-side heuristic simulation engine, enabling an interactive 3D visualization platform where researchers and the public alike can explore ‘what-if’ seismic scenarios in real time. The remainder of this

II. OBJECTIVES

1. To develop a robust machine-learning framework for forecasting earthquake magnitude and surface severity utilizing historical, geographical, and instrumental data.
2. To compare conventional algorithms with ensemble methods (Random Forest, XGBoost) for seismic regression and classification tasks using MSE, R^2, and F1 metrics.
3. To identify naturally occurring global tectonic risk zones using K-Means clustering on geospatial data.
4. To integrate real-time USGS GeoJSON telemetry into a dynamic front-end dashboard featuring a 3D Earth visualization.
5. To engineer an interactive heuristic simulator allowing users to manipulate geological parameters (fault type, depth, soil condition) to instantly visualize theoretical seismic destruction.

III. METHODOLOGY

A. Data Validation and Understanding: Inspect the USGS dataset structure, handling missing values, standardizing timestamps, and analyzing geographical variance across the Ring of Fire and Mid-Atlantic Ridge. This foundational step ensures that noise, duplication, and inconsistent timestamp formats in the raw records are identified early, before they can propagate into downstream feature engineering and model training.

B. Exploratory Data Analysis (EDA): Analyze depth distribution, magnitude frequency (Gutenberg-Richter law), and seasonal trends using descriptive statistics. These exploratory findings guide subsequent modelling decisions,

such as whether depth and magnitude should be treated as independent features or combined into interaction terms.

C. Data Cleaning and Feature Engineering: Encode spatial bounding boxes (lat/lon grids). Construct interaction features representing energy decay over depth. Map numeric magnitude to human-readable severity classes ('Low', 'Medium', 'High', 'Catastrophic'). Careful feature engineering at this stage is intended to give the downstream models physically meaningful signals rather than raw, unprocessed coordinates.

D. Regression Modelling: Compare Linear Regression, Random Forest, and Gradient Boosting (XGBoost) for next-event magnitude prediction based on spatial and temporal inputs. Comparing a simple linear baseline against ensemble methods allows the project to quantify how much predictive benefit is gained from added model complexity, rather than assuming ensembles are superior by default.

E. Classification Modelling: Predict the surface severity class using Random Forest classifiers evaluated via Precision, Recall, and F1-Score to handle class imbalances (severe earthquakes are rare). These metrics are prioritized over raw accuracy specifically because catastrophic events, while rare, are the cases the system most needs to identify correctly.

F. Unsupervised Analysis: Apply K-Means clustering on geographic coordinates to naturally partition the globe into high-risk tectonic zones without explicit labels. Because this clustering is performed without labels, the resulting zones serve as an independent, data-driven check on whether the model's implicit understanding of risk aligns with established tectonic boundaries.

G. Interactive Simulation Engine: Translate the learned weights of the ML models into a fast JavaScript-based heuristic engine for instantaneous client-side 'what-if' scenario testing, incorporating wave amplification physics (soil conditions) and inverse-square distance decay. By approximating the trained models' behaviour with lightweight heuristics, the simulator can respond to user input instantaneously in the browser, without a server round-trip for every parameter change.

H. Application Integration: Develop a premium web interface using HTML/CSS/JS, integrating Globe.gl for 3D visualization, Chart.js for analytical reporting, and Web Audio API for seismic sonification. Together, these components are intended to make seismic risk exploration accessible to non-specialist users, not just researchers familiar with raw model outputs.

IV. Data Source

The primary dataset is derived from the United States Geological Survey (USGS) Earthquake Hazards Program, containing comprehensive historical records of global seismic events. Important variables include latitude, longitude, depth, magnitude (mag), magnitude type (magType), number of reporting stations (nst), azimuthal gap (gap), root-mean-square travel time residual (rms), and minimum distance to the closest station (dmin). The dataset represents decades of telemetry and serves as the foundational truth for training both the regression and classification pipelines.

As the authoritative United States government agency responsible for monitoring natural hazards, the USGS provides continuously updated, quality-controlled seismic records that are widely used as a benchmark in academic seismology research, lending strong external validity to models trained on this dataset.

V. Plan of Execution

Stage	Planned Work
Project setup and data ingestion	Finalize problem definition, establish data pipelines from USGS CSV files, and set up the Python analytical environment.
Data understanding and EDA	Validate the dataset, handle missing variables, and conduct statistical analysis on depth vs. magnitude correlations.
Feature Engineering	Develop geospatial interaction features, depth categorization, and severity class mapping.
Model development	Train and compare Random Forest and XGBoost models for magnitude prediction and severity classification.
Unsupervised analysis	Perform K-Means clustering to map geographical risk zones and validate against known tectonic boundaries.
Backend Pipeline Generation	Develop Python scripts to automate inference and export JSON payloads for frontend consumption.

Stage	Planned Work
Application development	Build the dashboard interface, integrating Globe.gl for 3D mapping, Chart.js, and the live USGS polling ticker.
Simulator Implementation	Develop the JavaScript heuristic simulator allowing real-time parameter tweaking (fault type, soil condition).
Final validation and documentation	Consolidate results, prepare the project synopsis, and execute end-to-end system testing.

All modelling, evaluation, and application development activities are designed to be executed sequentially, culminating in a production-ready demonstration.

VI. Expected Outcomes

- A trained regression model capable of estimating the magnitude of future seismic events from geographic, temporal, and instrumental inputs.
- A classification model that maps predicted magnitude to a human-interpretable severity category (Low, Medium, High, Catastrophic).
- A set of data-driven tectonic risk zones derived from unsupervised clustering of historical earthquake locations.
- An interactive 3D WebGL globe that ingests real-time USGS telemetry and visualizes global seismic activity as it occurs.
- A client-side 'what-if' simulator that lets users vary fault mechanism, soil condition, and depth to explore theoretical seismic outcomes without re-running the underlying models.

VII. Conclusion

By combining ensemble machine learning, unsupervised clustering, and an interactive 3D visualization layer, ‘On Shaky Grounds’ aims to bridge the gap between raw seismic telemetry and actionable, intuitive risk communication. Rather than treating prediction and visualization as separate concerns, the project distils trained models into a responsive simulation engine, allowing researchers, students, and the public to interrogate seismic risk directly rather than through static reports alone. The objectives, methodology, and execution plan outlined in this synopsis describe an achievable, staged path toward that goal, culminating in a production-ready demonstration system.

VIII. References

[1] A. Majmudar et al., “Application of machine learning algorithms for earthquake magnitude prediction,” Journal of Seismology, vol. 27, pp. 112-128, 2025.

[2] R. B. Smith and M. Johnson, “A systematic review of AI in seismology,” Computational Geosciences, vol. 34, 10231, 2024.

[3] K. Reddy, “Forecasting seismic activity using Random Forest and XGBoost,” Artificial Intelligence in Earth Science, vol. 12, pp. 45-60, 2023.

[4] T. Nakagawa et al., “Deep learning approaches for real-time earthquake early warning systems,” IEEE Transactions on Geoscience, vol. 62, pp. 312-325, 2024.

[5] M. Rayyan and S. Kumar, “Interpretable Machine Learning Approaches for Tectonic Risk Assessment,” Earth and Space Science, vol. 11, 230151, 2024.

[6] J. Asencio et al., “Internet of Things and global seismic telemetry monitoring,” Engineering Applications of Artificial Intelligence, vol. 137, 109182, 2024.

[7] L. Chen and H. Wang, “Do ensemble methods improve the prediction of deep-focus earthquakes?,” Seismological Research Letters, vol. 95, 119751, 2024.

[8] S. K. Natarajan et al., “Optimized machine learning model for tectonic risk prediction in the Pacific Ring of Fire,” Scientific Reports, vol. 14, 6795, 2024.

[9] A. Sharma and V. Gupta, “Integrating machine learning techniques for seismic hazard mapping,” Earth Science Informatics, vol. 17, pp. 3733-3748, 2024.

[10] R. P. Kumar et al., “Machine learning-based prediction of hazard zones: a case study of the Himalayan Belt,” Discover Geoscience, vol. 2, art. 34, 2024.

[11] M. Sahu and A. Das, “Leveraging Satellite and USGS Telemetry for Predicting Megathrust Events,” Geological Data Research, vol. 24, 240066, 2024.

[12] H. Zhang et al., “Multimodal imputation-based stacked ensemble for seismic classification,” Computers & Geosciences, vol. 114, 109098, 2024.

[13] P. Rossi and G. Bianchi, “Statistical learning models for sustainable disaster management,” Intelligent Systems with Applications, vol. 22, 200333, 2024.

[14] G. Ravindiran et al., “Spatio-temporal seismic analysis over coastal regions using artificial intelligence,” Ecological Informatics, vol. 76, 102067, 2023.

[15] O. Kothari et al., “Forecasting Global Earthquakes: A Machine Learning Approach,” in 2023 4th International Conference on Intelligent Technologies (CONIT), IEEE, 2024.